

Redefining the Observable Universe in WRM

The radius of the observable universe in the Standard Model (Λ -CDM) (approximately 46.5 billion light-years) is the "range that light can travel," based on the expansion rate of the universe. However, in **the WRM (Spacetime Refraction Model)**, this distance is derived as **the "attenuation limit of light due to the refractive index gradient of spacetime."**

1. Basic principle: Energy dissipation process

In WRM, when light crosses the fundamental wave potential of the cosmic background, which is one trillion light-years away, the distance r The photon energy is proportional to E We consider that it dissipates into the "medium" of spacetime. This replaces the "expansion of space" in the Standard Model with **"redshift due to refraction of spacetime" in WRM.**

[Observation limit distance] R_{obs} [Derivation of]

Hubble constant H_0 And the WRM correction coefficient derived from the DE440 residual $\alpha = 0.046$ Using this, light completely loses energy (redshift). $z \rightarrow \infty$ The distance is approximated by the following formula:

$$R_{obs} \approx \frac{c}{H_0} \times \frac{1}{\sqrt{\alpha}}$$

Here, the standard Hubble distance $c/H_0 \approx 144$ Substituting billions of light-years:

$$R_{obs} \approx 144 \text{ hundred million light year} \times \frac{1}{\sqrt{0.046}} \approx 144 \times 4.66$$

$$R_{obs} \approx 670 \text{ billion light years}$$

2. Consistency of numerical interpretation

The figure of approximately 67 billion light-years derived from the WRM is larger than the 46.5 billion light-years of the Standard Model, suggesting a physical limit to how far the "echoes" of the cosmic microwave background radiation (CMB) can travel. This is because the fundamental wavelength is 1 trillion light-years. $\lambda \approx 10^{12}$ This corresponds to approximately 1/15th of LY and geometrically coincides with **the "extension of the first interference region" in wave structures.**

In conclusion, the observable universe in WRM refers not to the end of expansion, but to **"the boundary where the refractive index of spacetime can no longer preserve the information of light."** Beyond this boundary, a gigantic structure (the main body) extends for a trillion light-years, but our means of observation, "light," is limited by its refractive index and cannot see any deeper into the abyss.